5.0 ETHERNET OVERVIEW

Before discussing the use of the ENC28J60 as an Ethernet interface, it may be helpful to review the structure of a typical data frame. Users requiring more information should refer to the IEEE 802.3 standard which is the basis for the Ethernet protocol.

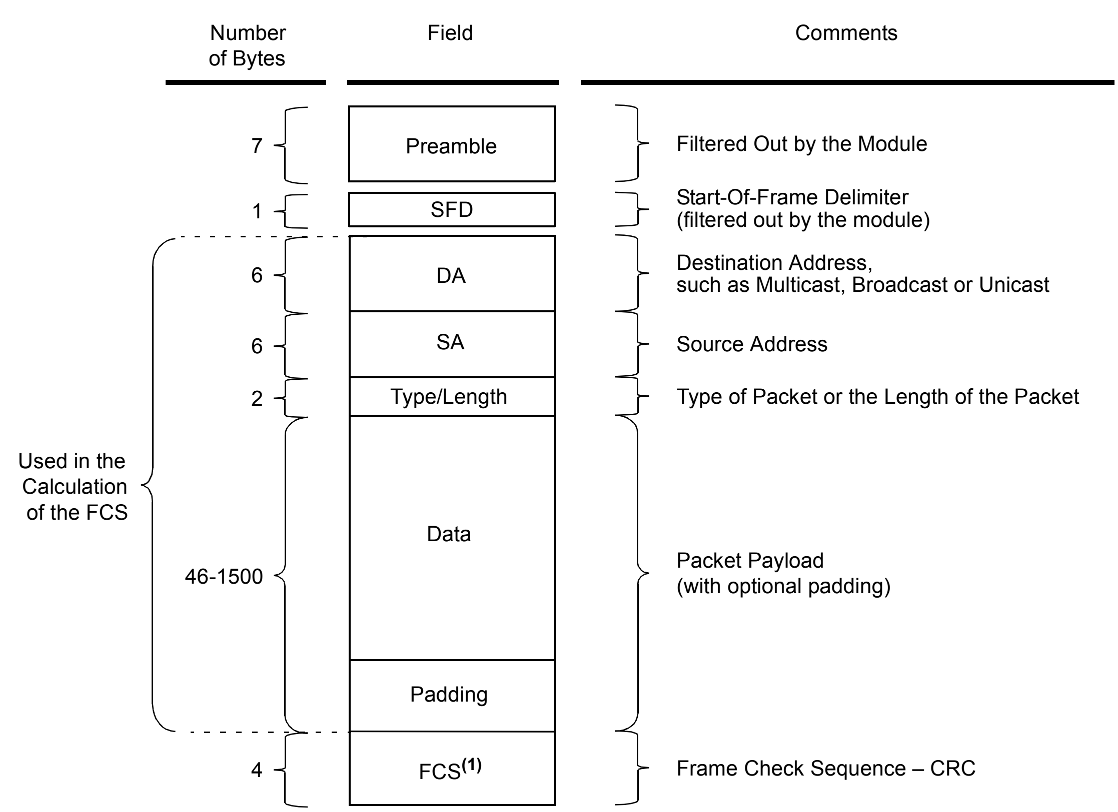
5.1 Packet Format

Normal IEEE 802.3 compliant Ethernet frames are between 64 and 1518 bytes long. They are made up of five or six different fields: a destination MAC address, a source MAC address, a type/length field, data payload, an optional padding field and a Cyclic Redundancy Check (CRC). Additionally, when transmitted on the Ethernet medium, a 7-byte preamble field and Start-Of-Frame (SOF) delimiter byte are appended to the beginning of the Ethernet packet. Thus, traffic seen on the twisted-pair cabling will appear as shown in Figure 5-1.

5.1.1 PREAMBLE/START-OF-FRAME DELIMITER

When transmitting and receiving data with the ENC28J60, the preamble and Start-Of-Frame delimiter bytes will automatically be generated or stripped from the packets when they are transmitted or received. The host controller does not need to concern itself with them. Normally, the host controller will also not need to concern itself with padding and the CRC which the ENC28J60 will also be able to automatically generate when transmitting and verify when receiving. The padding and CRC fields will, however, be written into the receive buffer when packets arrive, so they may be evaluated by the host controller if needed.

FIGURE 5-1: ETHERNET PACKET FORMAT



Note 1: The FCS is transmitted starting with bit 31 and ending with bit 0.

5.1.2 DESTINATION ADDRESS

The destination address field is a 6-byte field filled with the MAC address of the device that the packet is directed to. If the Least Significant bit in the first byte of the MAC address is set, the address is a Multicast destination. For example, 01-00-00-00-F0-00 and 33-45-67-89-AB-CD are Multicast addresses, while 00-00-00-00-F0-00 and 32-45-67-89-AB-CD are not.

Packets with Multicast destination addresses are designed to arrive and be important to a selected group of Ethernet nodes. If the destination address field is the reserved Multicast address, FF-FF-FF-FF-FF-FF, the packet is a Broadcast packet and it will be directed to everyone sharing the network. If the Least Significant bit in the first byte of the MAC address is clear, the address is a Unicast address and will be designed for usage by only the addressed node.

The ENC28J60 incorporates receive filters which can be used to discard or accept packets with Multicast, Broadcast and/or Unicast destination addresses. When transmitting packets, the host controller is responsible for writing the desired destination address into the transmit buffer.

5.1.3 SOURCE ADDRESS

The source address field is a 6-byte field filled with the MAC address of the node which created the Ethernet packet. Users of the ENC28J60 must generate a unique MAC address for each controller used.

MAC addresses consist of two portions. The first three bytes are known as the Organizationally Unique Identifier (OUI). OUIs are distributed by the IEEE. The last three bytes are address bytes at the discretion of the company that purchased the OUI.

When transmitting packets, the assigned source MAC address must be written into the transmit buffer by the host controller. The ENC28J60 will not automatically transmit the contents of the MAADR registers which are used for the Unicast receive filter.

5.1.4 TYPE/LENGTH

The type/length field is a 2-byte field which defines which protocol the following packet data belongs to. Alternately, if the field is filled with the contents of 05DCh (1500) or any smaller number, the field is considered a length field and it specifies the amount of non-padding data which follows in the data field. Users implementing proprietary networks may choose to treat this field as a length field, while applications implementing protocols such as the Internet Protocol (IP) or Address Resolution Protocol (ARP), should program this field with the appropriate type defined by the protocol's specification when transmitting packets.

5.1.5 DATA

The data field is a variable length field, anywhere from 0 to 1500 bytes. Larger data packets will violate Ethernet standards and will be dropped by most Ethernet nodes. The ENC28J60, however, is capable of transmitting and receiving larger packets when the Huge Frame Enable bit is set (MACON3.HFRMEN = `1`).

5.1.6 PADDING

The padding field is a variable length field added to meet IEEE 802.3 specification requirements when small data payloads are used. The destination, source, type, data and padding of an Ethernet packet must be no smaller than 60 bytes. Adding the required 4-byte CRC field, packets must be no smaller than 64 bytes. If the data field is less than 46 bytes long, a padding field is required.

When transmitting packets, the ENC28J60 automatically generates zero padding if the MACON3.PADCFG<2:0> bits are configured to do so. Otherwise, the host controller should manually add padding to the packet before transmitting it. The ENC28J60 will not prevent the transmission of undersize packets should the host controller command such an action.

When receiving packets, the ENC28J60 automatically rejects packets which are less than 18 bytes; it is assumed that a packet this small does not contain even the minimum of source and destination addresses, type information and FCS checksum required for all packets. All packets 18 bytes and larger will be subject to the standard receive filtering criteria and may be accepted as normal traffic. To conform with IEEE 802.3 requirements, the application itself will need to inspect all received packets and reject those smaller than 64 bytes.

5.1.7 CRC

The CRC field is a 4-byte field which contains an industry standard 32-bit CRC calculated with the data from the destination, source, type, data and padding fields.

When receiving packets, the ENC28J60 will check the CRC of each incoming packet. If ERXFCON.CRCEN is set, packets with invalid CRCs will automatically be discarded. If CRCEN is clear and the packet meets all other receive filtering criteria, the packet will be written into the receive buffer and the host controller will be able to determine if the CRC was valid by reading the receive status vector (see **Section 7.2 "Receiving Packets"**).

When transmitting packets, the ENC28J60 will automatically generate a valid CRC and transmit it if the MACON3.PADCFG<2:0> bits are configured to cause this. Otherwise, the host controller must generate the CRC and place it in the transmit buffer. Given the complexity of calculating a CRC, it is highly recommended that the PADCFG bits be configured such that the ENC28J60 will automatically generate the CRC field.

